Fiscal Year:	FY 2019	**Task Last Updated:**	FY 02/14/2019
PI Name:	Meyerowitz, Elliott Ph.D.		
Project Title:	Using GeneLab Data to Identify Novel Gravity Sensory Components in Arabidopsis		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91125-0001	**Congressional District:**	27
Comments:			
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System
Start Date:	01/01/2017	**End Date:**	12/31/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Altinok, Alphan Ph.D. (NASA Jet Propulsion Laboratory) Crichton, Daniel M.S. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	NNX17AD53G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Gravity is a critical regulator of plant directional growth patterns through reprogramming of gene expression patterns, and through defining the subcellular location of proteins such as the PINFORMED3 auxin efflux carrier. Plants have multiple sensory systems that respond to mechanical forces that include not only gravity, but also touch, and tension at the cell, tissue, and whole-organism level. Plant growth in space is affected by all of these – not just gravity directly, but also through the other modalities of mechanical sensing that are indirectly affected by gravity, for example, shoot bending and other changing tissue shapes. Mechanical stimuli have both short-term effects, such as creation of cellular calcium transients and tissue-wide waves over seconds and minutes, and long-term effects, such as reorganization of the microtubule cytoskeleton, and alteration of the subcellular location of proteins such as the auxin efflux carriers, that occur over 2-4 hours. The molecular components of gravitational perception mechanisms and of the short-term and long-term responses of plant cells and tissues to mechanical stimuli of all types are almost entirely unknown. The objective of the research was to use data from NASA's GeneLab to identify differentially expressed genes that could be responding to the mechanical changes that accompany space flight, and to use these genes to study plant growth responses to the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The broadened knowledge of plant growth mechanisms expected from these studies will be of direct benefit to humans traveling on extended space missions, where long-term plant growth is important for astronaut health and nutrition, and on Earth, for managing the effects of physical stresses on crop growth.
Task Progress:	We have used GenBank data sets and data from plant growth experiments on the International Space Station to identify classes of genes differentially regulated in the microgravity environment. We focused on three groups of genes – those that may act to mediate the auxin responses downstream of mechanically induced calcium fluxes, genes whose products may act in the calcium response itself, and transmembrane receptor kinases that may act in signaling cell wall information to plant cells. We also characterized, in detail, the calcium waves that result from mechanical stimulation of shoot meristems, and showed that the calcium response of meristems is necessary for the subsequent mechanical response of the auxin efflux protein that dictates auxin flow in plant stem cell populations. This broadened knowledge of plant growth mechanisms will be of direct benefit to humans traveling on extended space missions, where long-term plant growth is important for astronaut health and nutrition, and on Earth, for managing the effects of physical stresses on crop growth.
Bibliography Type:	Description: (Last Updated: 02/19/2019)
Articles in Peer-reviewed Journals	Li T, Yan A, Bhatia N, Altinok A, Afik E, Durand-Smet P, Tarr PT, Schroeder JI, Heisler MG, Meyerowitz EM. "Calcium signals are necessary to establish auxin transporter polarity in a plant stem cell niche." Nat Commun. 2019 Feb 13;10(1):726. https://doi.org/10.1038/s41467-019-08575-6 ; PubMed PMID: 30760714; PubMed Central PMCID: PMC6374474 , Feb-2019